Homework 3

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MATH 421 – The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used, except for the textbook *Elementary Analysis: The Theory of Calculus* by Ross and my own class notes.

Problem 1

Prove if x and y are real numbers, then:

A. $|x-y| \leq |x| + |y|$

B. $|x| - |y| \leq |x-y|$

Proof.

A. We want to prove $|x-y| \leq |x| + |y|$.

First write the difference as a sum:

$$x - y = x + (-y).$$

By the triangle inequality applied to the numbers x and $-y$,

$$|x-y| = |x + (-y)| \leq |x| + |-y| = |x| + |y|.$$

Thus $|x-y| \leq |x| + |y|$, as required.

B. We want to prove $|x| - |y| \leq |x-y|$.

Start with the triangle inequality for the sum $(x-y)+y$:

$$|(x-y)+y| \leq |x-y| + |y|.$$

The left-hand side simplifies to $|x|$, so

$$|x| \leq |x-y| + |y|.$$

Subtract $|y|$ from both sides (this is valid because $|y|$ is a real number):

$$|x| - |y| \leq |x-y|.$$

Hence $|x| - |y| \leq |x-y|$, as desired.

□

Problem 2 (Ross Chapter 1, 3.6)

a. Prove $|a+b+c| \leq |a| + |b| + |c|$ for all $a, b, c \in \mathbb{R}$.
Hint: Apply the triangle inequality twice. Do not consider eight cases.

Proof. By associativity of addition,

$$|a+b+c| \;=\; |(a+b)+c|.$$

Applying the triangle inequality $|u+v| \leq |u| + |v|$ with $u = a+b$ and $v = c$ gives

$$|(a+b)+c| \;\leq\; |a+b| + |c|.$$

Apply the triangle inequality again to $|a+b|$:

$$|a+b| \;\leq\; |a| + |b|.$$

Substituting this into the previous inequality yields

$$|a+b+c| \;\leq\; (|a| + |b|) + |c| \;=\; |a| + |b| + |c|,$$

as required. □

b. Use induction to prove that for n real numbers $a_1, \ldots, a_n$,

$$\big|a_1 + a_2 + \cdots + a_n\big| \;\leq\; |a_1| + |a_2| + \cdots + |a_n|.$$

Proof. Let $P(n)$ denote the statement

$$\big|a_1 + \cdots + a_n\big| \leq |a_1| + \cdots + |a_n| \quad \text{for all real } a_1, \ldots, a_n.$$

Base case. For $n = 1$, $|a_1| \leq |a_1|$ is trivial. (If you prefer to start at $n = 2$, then $|a_1 + a_2| \leq |a_1| + |a_2|$ is exactly the triangle inequality.)

Inductive step. Assume $P(n)$ holds for some $n \geq 1$; i.e.,

$$\big|a_1 + \cdots + a_n\big| \leq |a_1| + \cdots + |a_n| \quad \text{for all } a_1, \ldots, a_n \in \mathbb{R}.$$

Let $a_1, \ldots, a_n, a_{n+1} \in \mathbb{R}$ be arbitrary. Using associativity and the triangle inequality,

$$\begin{aligned}\big|a_1 + \cdots + a_n + a_{n+1}\big| &= \big|(a_1 + \cdots + a_n) + a_{n+1}\big| \\ &\le \big|a_1 + \cdots + a_n\big| + |a_{n+1}|.\end{aligned}$$

By the inductive hypothesis,

$$\big|a_1 + \cdots + a_n\big| \le |a_1| + \cdots + |a_n|.$$

Combining the two inequalities gives

$$\big|a_1 + \cdots + a_n + a_{n+1}\big| \le (|a_1| + \cdots + |a_n|) + |a_{n+1}| = |a_1| + \cdots + |a_n| + |a_{n+1}|.$$

Thus $P(n+1)$ holds.

By the principle of mathematical induction, $P(n)$ is true for every $n \in \mathbb{N}$. □

Problem 3 (Ross Chapter 1, 3.8)

Let $a, b \in \mathbb{R}$. Show: if $a \le b_1$ for every $b_1 > b$, then $a \le b$.

Proof. We prove the contrapositive. Suppose $a > b$. Define

$$b_1 = \frac{a+b}{2}.$$

Since $a > b$, we have $a - b > 0$, hence $\dfrac{a-b}{2} > 0$. Adding b to both sides gives

$$b + \frac{a-b}{2} = \frac{a+b}{2} = b_1 \ > \ b.$$

Also, from $a - b > 0$ we get $a - \dfrac{a-b}{2} = \dfrac{a+b}{2} = b_1 < a$. Therefore $b_1 > b$ and $a \not\le b_1$.

Thus, if $a > b$ there exists $b_1 > b$ for which $a \not\le b_1$. This is the contrapositive of the desired implication, and hence the original statement holds: if $a \le b_1$ for every $b_1 > b$, then $a \le b$. □

Problem 4

Definition. An integer n is divisible by an integer d if $n = dk$ for some $k \in \mathbb{Z}$.

Theorem. Suppose n is an integer. If n^2 is divisible by 3, then n is divisible by 3.

Proof. We proceed by proving the contrapositive: If n is not divisible by 3, then n^2 is not divisible by 3.

If n is not divisible by 3, then by the Division Algorithm we can write

$$n = 3k + 1 \quad \text{or} \quad n = 3k + 2$$

for some integer k.

Case 1. If $n = 3k + 1$, then

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1.$$

Thus $n^2 \equiv 1 \pmod 3$, so n^2 is not divisible by 3.

Case 2. If $n = 3k + 2$, then

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1.$$

Again, $n^2 \equiv 1 \pmod 3$, so n^2 is not divisible by 3.

In both cases, if n is not divisible by 3, then n^2 is not divisible by 3. By contrapositive, if n^2 is divisible by 3, then n must be divisible by 3. □

Problem 5

Prove that $\sqrt{3}$ is irrational.

Proof. Suppose, toward a contradiction, that $\sqrt{3}$ is rational. Then there exist integers p, q with $q \neq 0$ such that

$$\sqrt{3} = \frac{p}{q} \qquad \text{and} \qquad \gcd(p, q) = 1$$

(i.e., the fraction is in lowest terms). Squaring both sides yields

$$3 = \frac{p^2}{q^2} \quad \Longrightarrow \quad p^2 = 3q^2.$$

Hence p^2 is divisible by 3. By the result of Problem 4 (if n^2 is divisible by 3, then n is divisible by 3), it follows that $3 \mid p$. Thus there exists $k \in \mathbb{Z}$ with $p = 3k$. Substituting back, we obtain

$$3q^2 = p^2 = (3k)^2 = 9k^2 \quad \Longrightarrow \quad q^2 = 3k^2,$$

so $3 \mid q$ as well.

Therefore p and q are both divisible by 3, which contradicts $\gcd(p, q) = 1$. This contradiction shows our initial assumption was false. Hence $\sqrt{3}$ is irrational. □

Problem 6

Prove that $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$ are both irrational numbers.

Proof. **(1) $\boldsymbol{\sqrt{2} + \sqrt{3}}$ is irrational.** Suppose, for a contradiction, that $r = \sqrt{2} + \sqrt{3} \in \mathbb{Q}$. Then $\sqrt{3} = r - \sqrt{2}$ and squaring gives

$$3 = (r - \sqrt{2})^2 = r^2 - 2r\sqrt{2} + 2.$$

Rearranging yields

$$-2r\sqrt{2} = 1 - r^2 \qquad \Longrightarrow \qquad \sqrt{2} = \frac{r^2 - 1}{2r}.$$

Since $r \in \mathbb{Q}$, the right-hand side is rational, which implies $\sqrt{2} \in \mathbb{Q}$, a contradiction. Hence $\sqrt{2} + \sqrt{3}$ is irrational.

(2) $\boldsymbol{\sqrt{2} - \sqrt{3}}$ is irrational. Observe that

$$(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) = 2 - 3 = -1.$$

If $\sqrt{2} - \sqrt{3}$ were rational, then $\sqrt{2} + \sqrt{3} = -1/(\sqrt{2} - \sqrt{3})$ would be rational, contradicting part (1). Therefore $\sqrt{2} - \sqrt{3}$ is irrational. $\square$